

2025 Urban Mobility Report

Appendix C: Value of Delay Time for Use in Mobility Monitoring Efforts

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SUMMARY

The value of delay time for passenger vehicle motorists uses the median hourly wage rate for all occupations as produced by the Bureau of Labor Statistics (BLS) as a base**. Researchers estimate the 2024 value of delay time for personal travel at $24.01 per person and $36.02 per personal vehicle. The commercial value of travel time is now based on the American Transportation Research Institute (ATRI) annual survey modified by speed, type of vehicle, and vehicle occupancy and is estimated to be $80.16 per vehicle per hour for 2024. Neither the value of delay time for personal nor commercial vehicles include the cost of fuel.**

INTRODUCTION

The value of delay time is an estimate of the average differential cost of the extra travel time resulting from congestion. As it relates to the *Urban Mobility Report* (*UMR*) methodology, this congestion cost is a function of both the time and fuel used while the motorist is in congested traffic. The cost of extra fuel consumed during congestion is computed separately from the time spent (wasted) in congestion—fuel cost is not a subject of this report. This report focuses on the computation of an updated value of delay time for passenger cars and trucks. This value of delay time serves as an input to compute urban area congestion cost from urban area delay.

For passenger car motorists, this value of delay time is based on hourly wage rates. For truck drivers, the value of delay time is expressed as the wage rate of the driver multiplied by truck occupancy plus the various operating cost components associated with a straight truck or tractor-trailer. This report summarizes the components of each value of delay time cost estimate and the updated values of time used in the UMR calculations beginning with 2017 data.

METHODOLOGY FOR PASSENGER VEHICLE MOTORIST VALUE OF DELAY TIME

The value of vehicle occupant time is based on median BLS wage estimates for all occupations. The median hourly wage for 2024 has not yet been released. However, the BLS estimates that wages increased 3.9 percent for the 12-month period ending in September 2024. Applying this percentage to the published 2023 median data, the estimate for 2024 would be $24.01 on a per person basis and yields an estimated value of delay time of $36.02 per personal vehicle using a vehicle occupancy rate of 1.5 persons per vehicle.

In earlier iterations of value of delay time calculations, a speed choice model developed by Chui and McFarland of the Texas Transportation Institute (TTI) was used by the Texas Department of Transportation *(1)*. The research indicated the value of

delay time was $11.98 in 1997 in Texas and was consistent at the time with estimates produced by other states. Using this methodology, Exhibit C-1 takes the $11.98 value of delay time found in 1997 and adjusts it for inflation by the Consumer Price Index (CPI) back to 1982 as well as forward to 2016.

There has long been a concern that simply taking the perceived value of delay time from an earlier study and adjusting it forward and backward by the CPI might prove to be problematic over time in that the CPI (a rate of change based on a market basket of goods for all urban consumers) may not be reflective of the actual value of passenger vehicle travel.

Beginning in 2016, the decision was made to use data published in the Occupational Employment Statistics series by the Bureau of Labor Statistics that provide both a mean and median hourly wage for all job classifications taken together *(2)*. The median hourly wage was chosen for use in this study because the median value eliminates the effect of extremes at either end of the wage range.

Exhibit C-1. Value of Passenger Vehicle Motorist Time

Year	Personal Value of Time	Consumer Price Index (1982–1984 Base Year)	Bureau of Labor Statistics Median Hourly Wage
1980	$6.15	82.4	–
1981	$6.78	90.9	–
1982	$7.20	96.5	–
1983	$7.43	99.6	–
1984	$7.75	103.9	–
1985	$8.03	107.6	–
1986	$8.18	109.6	–
1987	$8.48	113.6	–
1988	$8.83	118.3	–
1989	$9.25	124.0	–
1990	$9.75	130.7	–
1991	$10.17	136.2	–
1992	$10.47	140.3	–
1993	$10.78	144.5	–
1994	$11.06	148.2	–
1995	$11.37	152.4	–
1996	$11.71	156.9	–
1997	$11.98	160.5	–
1998	$12.17	163.0	–
1999	$12.43	166.6	–

Year	Personal Value of Time	Consumer Price Index (1982–1984 Base Year)	Bureau of Labor Statistics Median Hourly Wage
2000	$12.85	172.2	—
2001	$13.22	177.1	—
2002	$13.43	179.9	—
2003	$13.73	184.0	$13.53
2004	$14.10	188.9	$13.83
2005	$14.58	195.3	$14.15
2006	$15.06	201.8	$14.61
2007	$15.47	207.3	$15.10
2008	$16.07	215.3	$15.57
2009	$16.01	214.5	$15.95
2010	$16.28	218.1	$16.27
2011	$16.79	224.9	$16.57
2012	$17.14	229.6	$16.71
2013	$17.39	233.0	$16.87
2014	$17.67	236.7	$17.09
2015	$17.69	237.0	$17.40
2016	$17.91	240.0	$17.81
2017	—	—	$18.12
2018	—	—	$18.71
2019	—	—	$19.14
2020	—	—	$19.64
2021	—	—	$22.00
2022	—	—	$22.26
2023	—	—	$23.11
2024	—	—	$24.01

Note: — = Not applicable.

The annual *Your Driving Costs* report produced by the American Automobile Association was used as a basis to calculate the marginal cost per mile of travel for passenger vehicles *(3)*. The individual costs associated with the different classes of vehicles were weighted by vehicle production shares to produce a fleetwide average that represents the vehicle fleet currently in use. An average of 15,000 miles per year was used as the default annual mileage. Those data yielded an operating cost estimate of $0.819 per mile, as shown in Exhibit C-2. Excluding fuel, the passenger vehicle operating cost is $0.673 per mile. All the cost categories saw a slight decrease from 2023 to 2024 except for maintenance, repairs, and tire costs. The most notable year-to-year change is the rise in depreciation costs.

Exhibit C-2. 2024 Passenger Vehicle Operating Costs per Mile

Estimated Cost per Mile	2023 Average Cost	2024 Average Cost
Fuel	0.158	0.145
Maintenance, Repair, Tires	0.148	0.130
Insurance	0.099	0.108
License, Registration, Taxes	0.064	0.052
Depreciation	0.359	0.305
Finance Charges	0.099	0.079
Total	**0.926**	**0.819**
Total minus Fuel Cost	**0.769**	**0.673**

Source: American Automobile Association (3)

METHODOLOGY FOR TRUCK DRIVER VALUE OF DELAY TIME

ATRI conducts an annual survey of its membership to determine estimates of operational trucking costs *(4)*. TTI has closely followed ATRI's survey and has determined it currently provides the most accurate data available for commercial truck operating costs and should serve as the basis for the truck value of delay time estimate used in the UMR.

The ATRI survey disaggregates variable costs into nine categories:

- Fuel.
- Lease/purchase payments.
- Repairs and maintenance.
- Insurance.
- Permits and licenses.
- Tires.
- Tolls.
- Driver wages and benefits.

(For purposes of this report, the U.S. average price, as published by the Energy Information Administration, U.S. Department of Energy, were substituted for the fuel costs included in the ATRI survey.) Values are calculated on a per-mile and per-hour basis. Exhibit C-3 provides a summary of the survey results expressed on a per-mile basis for the period 2015–2021 published by ATRI and TTI estimates for 2022–2024. The 2024 estimated average operating cost for commercial trucks is $1.357 per mile. Excluding fuel, the truck operating cost per mile comes to $0.788 per mile.

Exhibit C-3. Estimates of Truck Operating Costs per Mile, 2015–2024

Estimated Cost Per Mile	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Fuel	0.403	0.357	0.410	0.422	0.392	0.321	0.461	0.599	0.529	0.569
Truck/Trailer Lease or Purchase Payments	0.230	0.255	0.264	0.265	0.256	0.271	0.279	0.287	0.343	0.378
Repair and Maintenance	0.156	0.166	0.167	0.171	0.149	0.148	0.175	0.180	0.203	0.212
Truck Insurance Premiums	0.074	0.075	0.075	0.084	0.071	0.087	0.086	0.089	0.091	0.104
Permits and Licenses	0.019	0.022	0.023	0.024	0.020	0.016	0.016	0.016	0.016	0.009
Tires	0.043	0.035	0.038	0.038	0.039	0.043	0.041	0.042	0.047	0.048
Tolls	0.020	0.024	0.027	0.030	0.035	0.037	0.032	0.033	0.029	0.036
Total	**0.945**	**0.934**	**1.004**	**1.034**	**0.962**	**0.923**	**1.090**	**1.247**	**1.259**	**1.357**
Total minus Fuel	**0.542**	**0.577**	**0.594**	**0.612**	**0.570**	**0.602**	**0.629**	**0.648**	**0.729**	**0.788**

Sources: American Transportation Research Institute (2013–2021) (4) and Texas A&M Transportation Institute (2022–2024).
Note: ATRI data from 2013–2023 are used in the calculation. Only data from 2015 forward are shown here due to space limitations.

Researchers used the ATRI estimates as a basis for calculating the 2024 commercial truck value of delay time. To do so, it was necessary to make several adjustments to the ATRI estimates to update them to 2024. Those methodological adjustments are as follows:

1. Segregated non-fuel and non-labor marginal costs from other costs.
2. Subtracted fuel costs.
3. Calculated the mean and median average percent increase in non-fuel, non-labor marginal costs.
4. Selected the median average percent increase in non-fuel, non-labor marginal cost and applied the increase to the ATRI 2023 non-fuel, non-labor costs to determine cost-per-mile operating costs for 2024.
5. Determined an estimate of fuel efficiency (miles/gallon) for tractor-trailer and straight trucks.
6. Determined an estimate of percent of commercial trucks that are tractor-trailer trucks and straight trucks.
7. Determined a labor cost estimate.

8. Estimated the cost of driver benefits.
9. Estimated an average speed for tractor-trailer trucks and straight trucks.

In the ATRI survey, the mean annual increase from 2013 through 2023 in non-fuel, non-labor costs per mile was calculated at 5.1 percent, and the median annual increase was calculated to be 5.0 percent. The annual percent change ranged from −6.9 percent to +15.0 percent. Given the wide range in annual percent change, the median value was used to calculate an estimate for non-fuel, non-labor cost-per-mile estimate for 2024 because the median is less impacted by outliers in the data.

As noted above, several additional estimates were made based on conversations with industry members and the experience of research professionals to produce a 2024 estimate for value of truck delay time. They are as follows:

- Estimate of percent of trucks by type:
  - Tractor-trailer trucks: 60 percent.
  - Straight trucks: 40 percent.
- Estimate of truck vehicle miles traveled by type
  - Tractor-trailer trucks: 77.5 percent.
  - Straight trucks: 22.5 percent.
- Estimate of average truck occupancy:
  - Tractor-trailer trucks: 1.1 persons per truck.
  - Straight trucks: 1.2 persons per truck.
  - Weighted average: 1.14 persons per truck.
- Average vehicle speed:
  - Tractor-trailer trucks: 45 miles per hour.
  - Straight trucks: 30 miles per hour.
  - Weighted average: 39 miles per hour.

Using the calculation procedures outlined above, these data yielded an estimated operational truck value of $1.357 per mile, as shown in Exhibit C-3.

Finally, researchers obtained 2023 driver wages from the National Occupational Employment and Wage Estimates for the United States published by the U.S. Department of Labor, BLS *(2)*. The median hourly driver wage reported by BLS for a heavy truck or tractor-trailer truck driver was reported to be $26.12, with the median driver wage for a light or delivery truck drive reported as $20.42 per hour. Weighting the wages by the distribution of trucks by type produces a blended wage rate of $23.84 per hour. However, using previously reported driver wage rates produced by the ATRI surveys from 2013 to 2023, calculating the median annual change in wage rate of 7.1 percent, and then applying that increase to the 2023 ATRI wage rate

produced an estimate for 2024 of $33.55 per hour. Given the disparity between the BLS and ATRI estimates, the ATRI wage rate was selected for use in the 2024 calculation. The basis for this decision was two-fold. First, it is believed the ATRI survey more accurately reflects market conditions given the shortage of truck drivers, and, second, it provides the most recent data available.

The mean annual percentage of the cost of driver benefits as a percent of driver wages was calculated to be 28.3 percent; the median was calculated to be 29.3 percent for the period 2013 to 2023. The annual percent change ranged from 24.2 percent to 30.9 percent. Again, the median value of 29.3 percent was chosen as an estimate of the cost of driver benefits as a percentage of driver wages because it is less affected by outliers in the data. Using the median percent change, driver benefits were estimated to be $9.82 for 2024. The combined cost of driver wages and benefits was multiplied by an average weighted occupancy of 1.14 persons per vehicle to produce an estimate of $49.43 per hour of delay time for commercial trucks.

Taken together, these estimates produced a 2024 value of delay time for truck drivers of $1.97 per mile for use in the UMR urban area statistics compared to the 2023 TTI estimate of $1.82 per mile, as seen in Exhibit C-4. Again, these per mile costs **do not** include fuel.

Exhibit C-4. Estimates of 2024 Truck Value of Delay Time in Dollars per Mile

Cost Per Mile	TTI Estimate 2023	TTI Estimate 2024
Truck/Trailer Lease or Purchase Payments	0.343	0.378
Repairs and Maintenance	0.203	0.212
Truck Insurance Premiums	0.091	0.104
Permits and Licenses	0.016	0.009
Tires	0.047	0.048
Tolls	0.029	0.036
Subtotal	**$0.729**	**$0.788**
Driver Wages	0.869	0.951
Driver Benefits	0.220	0230
Subtotal	**$1.089**	**$1.181**
Total	**$1.818**	**$1.969**

Note: The estimates for driver wages and benefits include an average vehicle occupancy of 1.14 persons per vehicle.

Exhibit C-5 uses the per mile calculation multiplied by the weighted speed (39 miles per hour) to produce an hourly estimate of commercial truck value of delay time of

$80.16 for 2024 compared to the TTI estimate of $73.95 per hour for 2023. These estimates for driver wages and benefits also include an average vehicle occupancy of 1.14 persons per vehicle.

Exhibit C-5. Estimates of 2024 Truck Value of Delay Time in Dollars per Hour

Cost Per Hour	TTI Estimate 2023	TTI Estimate 2024
Truck/Trailer Lease or Purchase Payments	$13.39	$14.75
Repairs and Maintenance	$7.93	$8.28
Truck Insurance Premiums	$3.56	$4.06
Permits and Licenses	$0.61	$0.37
Tires	$1.82	$1.88
Tolls	$1.13	$1.39
Subtotal	**$28.45**	**$30.73**
Driver Wages	$35.06	$38.24
Driver Benefits	$10.45	$11.19
Subtotal	**$45.51**	**$49.43**
Total	**$73.95**	**$80.16**

Note: The estimates for driver wages and benefits include an average vehicle occupancy of 1.14 persons per vehicle.

Exhibit C-6 shows the estimated historical truck value of delay time in dollars per hour. ATRI and TTI values for truck value of delay time are available from 2008 to 2024. The average annual growth rate of 2008 to 2018 was calculated and applied to years 2000 through 2007. For the years 1980 to 1999, the average annual growth rate was adjusted to account for the effects of deregulation. During that time period, the average operating cost per mile was estimated to have dropped 35 to 75 percent. This combined with the impacts of inflation produces a slower rate of growth for those years. Exhibit C-7 depicts the truck value of delay time from 1980 to 2024. The 2020 through 2024 estimates include an average vehicle occupancy of 1.14 persons per vehicle.

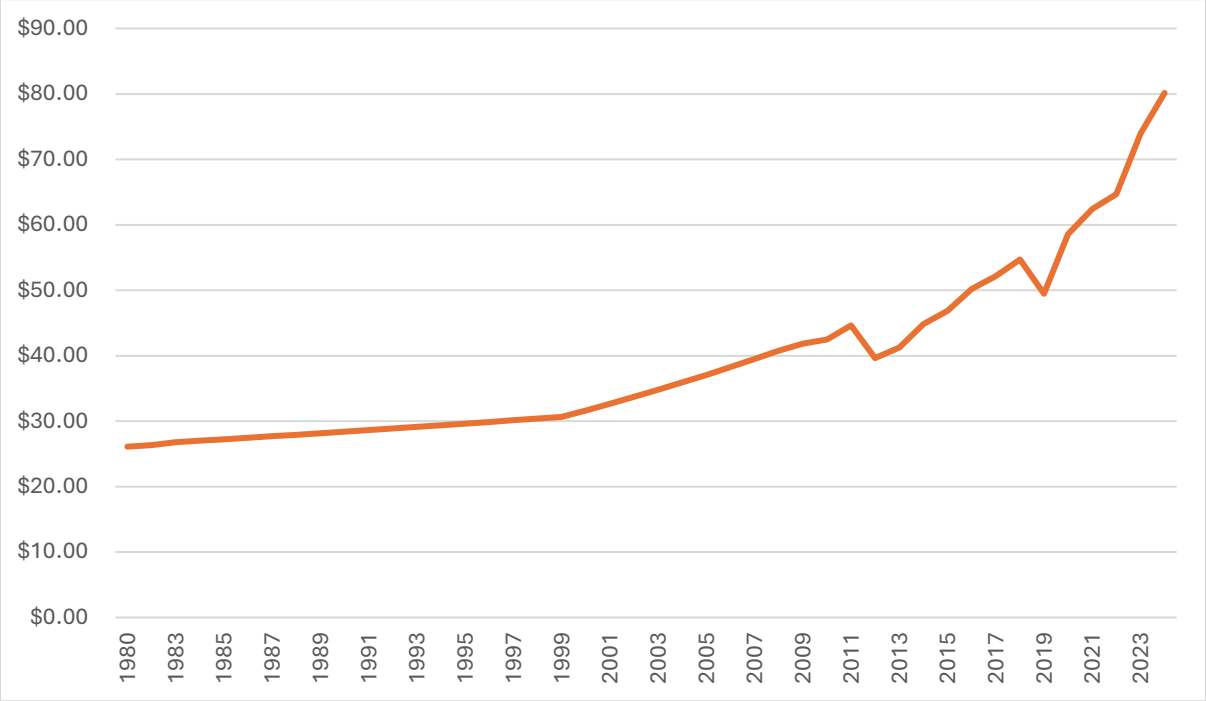
Exhibit C-6. Estimated Truck Value of Time in Dollars per Hour, 1980–2024

	Year	Estimated Truck Value of Time in Dollars per Hour
Average Annual Growth Rate 2008-2018 Adjusted to Account for Impacts of Deregulation	1980	$26.10
	1981	$26.33
	1982	$26.55
	1983	$26.77
	1984	$27.00
	1985	$27.23
	1986	$27.46
	1987	$27.69
	1988	$27.93
	1989	$28.17
	1990	$28.40
	1991	$28.65
	1992	$28.89
	1993	$29.13
	1994	$29.38
	1995	$29.63
	1996	$29.88
	1997	$30.13
	1998	$30.39
	1999	$30.65
Average Annual Growth Rate 2008-2018	2000	$31.64
	2001	$32.65
	2002	$33.71
	2003	$34.79
	2004	$35.91
	2005	$37.07
	2006	$38.26
	2007	$39.50

	Year	Estimated Truck Value of Time in Dollars per Hour
ATRI	2008	$40.77
	2009	$41.83
	2010	$42.50
	2011	$44.62
	2012	$39.66
	2013	$41.23
	2014	$44.82
	2015	$46.87
	2016	$50.20
	2017	$52.14
	2018	$54.71
	2019	$49.49
	2020	$58.54
	2021	$62.43
TTI	2022	$64.68
	2023	$73.95
	2024	$80.16

Note: 2020–2024 include 1.14 vehicle occupancy.

Exhibit C-7. Estimated Truck Value of Delay Time in Dollars per Hour, 1980–2024



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